

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

Evaluation from human preferences remains difficult across generative modeling settings because automatic metrics often capture only part of perceived quality. Human judgments are broadly informative but costly and noisy, with reliability varying across expertise, attention, display conditions, and diligence. Pairwise comparisons offer a practical evaluation primitive, yet converting incomplete and heterogeneous comparisons into global quality scores requires an aggregation model that can generalize beyond individual judgments. Bradley-Terry-based methods provide this foundation, but common variants either treat all raters as equally reliable or rely on optimization procedures without convergence guarantees, limiting their robustness and interpretability in broader evaluation pipelines. We introduce BBQ, a Bayesian Bradley-Terry variant that explicitly models rater quality, downweights or removes unreliable participants, and provides guaranteed monotonic likelihood convergence through an Expectation-Maximization algorithm. Across language and image-compression preference datasets, empirical results show that BBQ achieves faster convergence, well-calibrated uncertainty estimates, and more robust, interpretable rankings than baseline Bradley-Terry models, including in noisy or crowdsourced rater populations. The resulting framework supports more reliable and cost-effective aggregation of human pairwise comparisons for generative model evaluation and related preference-ranking tasks.

1 INTRODUCTION

Evaluating generative models is challenging across modalities, including large language models (LLMs) and image generators, because standard metrics often summarize properties that only partially align with human preferences. Metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, provide useful but incomplete signals about perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Consequently, human evaluations remain a central tool for establishing meaningful rankings whenever the target notion of quality is perceptual, subjective, or task-dependent. Leaderboards and comparative studies for large language models and Learned Image Compression (LIC) methods therefore rely heavily on human preference data rather than automatic scores alone.

Human evaluations, however, are expensive and time-consuming, so evaluation protocols must use limited judgments efficiently while controlling subjectivity and noise. Pairwise comparisons, where participants choose between two items rather than assigning absolute scores, provide a broadly useful and practical interface for such studies (Zerman et al., 2018; Wang et al., 2023). They are often easier for participants and can yield more consistent judgments across tasks and domains. At the same time, collecting every pair is rarely feasible because the number of possible comparisons grows quadratically with the number of items. When only a sparse subset of comparisons is observed, simple summaries such as win rates do not provide general item quality scores, because each win rate depends on which opponents were sampled.

To turn pairwise judgments into a ranking suitable for leaderboards or comparative studies, the comparisons must be aggregated by a model that can infer global scores from uneven local evidence. This requirement motivates robust and efficient aggregation methods that apply across evaluation

settings with different items, raters, and sampling designs. The Bradley-Terry (BT) (Bradley & Terry, 1952) model addresses this problem by estimating item scores from comparison outcomes. It has consequently become a standard probabilistic tool for aggregating pairwise judgments in many domains.

Despite this flexibility, the standard Bradley-Terry model leaves important practical issues unresolved. BT parameters can be estimated by iterative scaling methods (Ford Jr, 1957), gradient descent, or the minorization maximization (MM) algorithm (Hunter, 2004). For the basic BT model, MM algorithms guarantee convergence to the global maximum, whereas gradient-based methods, closely related to the Elo rating system, are widely deployed but lack the same guarantees (Hunter, 2004). When BT is extended to richer comparison settings, such as home field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may converge only to local optima (Hunter, 2004). An additional and broadly relevant source of difficulty is rater heterogeneity: participants may rush, guess, misunderstand the task, or lose focus, producing reliability that varies substantially across raters. If aggregation treats this variability as absent, unreliable judgments can distort item scores and make rankings less stable across repeated studies.

We propose a Bayesian Bradley-Terry variant that jointly models item quality and rater reliability for pairwise-comparison datasets with heterogeneous annotators. To the best of our knowledge, no prior work combines the BT framework with a Bayesian formulation that explicitly models rater quality while providing closed-form EM updates. Existing approaches that account for noisy raters typically rely on gradient-based optimization without convergence guarantees, whereas our method preserves stable convergence and yields interpretable item and rater parameters. The Bayesian formulation supplies regularization and a principled representation of epistemic uncertainty. To capture participant variability, we introduce rater-specific parameters that quantify how consistently each participant follows the underlying BT preference model, allowing the aggregation procedure to adapt the influence of individual comparisons. We derive an EM algorithm that estimates item and rater parameters efficiently through a latent-variable formulation, with guaranteed monotonic likelihood improvement. This design lets partially reliable raters contribute useful evidence while limiting the effect of inattentive or inconsistent participants. Because it models both item scores and rater quality, the method also provides uncertainty estimates that make rankings more interpretable and comparisons across studies more robust.

We evaluate the approach on several human preference datasets and introduce a newly collected *inhomogeneous rater quality* (IHQ) dataset¹. Within these language and image-compression benchmarks, the method shows faster convergence, greater robustness to noisy comparisons, and more consistent rankings than standard BT and naive aggregation methods. More broadly, the same modeling assumptions apply whenever noisy pairwise comparisons must be converted into reliable global rankings under limited annotation budgets. Overall, our work contributes a practical route toward more cost-efficient, interpretable, and reproducible human studies for evaluating AI-generated content and other preference-ranked items.

2 RELATED WORK

Many human studies ultimately require ranking a set of items, whether the items are models, outputs, products, or experimental conditions. Presenting participants with a choice between two alternatives rather than a single absolute score can increase sensitivity to subtle differences and reduce response variability (Zerman et al., 2018; Wang et al., 2023). For this reason, pairwise comparisons have become a common evaluation primitive for machine learning leaderboards and other preference-based benchmarks. In these settings, individual preferences between alternatives are aggregated into global rankings that support comparison across systems. Notable examples include the Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025), both of which collect pairwise comparisons and aggregate them using variants of the Bradley-Terry (BT) model.

The Bradley-Terry model (Bradley & Terry, 1952), first developed for ranking sports competitors, offers a probabilistic account of the chance that one item is preferred over another. By mapping local wins and losses to global item scores, it provides a general foundation for ranking from pair-

¹The IHQ dataset will be released upon acceptance.

wise evidence. This foundation has been used across games, consumer-preference studies, and other domains where comparative judgments are easier to collect than absolute measurements. A large family of extensions broadens the model to additional feedback structures and application assumptions. For example, the Plackett-Luce model generalizes BT from pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959), defining a probability distribution over permutations by multiplying successive BT probabilities (Luce et al., 1959). Other variants incorporate setting-specific factors, including home-field advantages (Agresti, 2010), ties (Rao & Kupper, 1967), or comparisons between groups rather than individuals (Huang et al., 2006).

Maximum likelihood estimation (MLE) remains a common route for inferring parameters in the basic BT model. Zermelo (1929) introduced an iterative estimator that has since become a standard computational approach. Lange et al. (2000) later showed that this procedure is an instance of minorization-maximization (MM), a class of algorithms that optimize surrogate functions to improve the likelihood. Hunter (2004) extended MM methods to generalized BT models and characterized when convergence to the MLE is guaranteed. For the classical BT model, convexity ensures convergence to the global optimum (Hunter, 2004). For richer model classes, including home-field advantage, multiple comparisons, or ties, the same optimization family may only guarantee convergence to a local maximum (Hunter, 2004).

Bayesian BT formulations extend this ranking framework by incorporating prior knowledge, regularization, and uncertainty quantification (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) showed that MM algorithms can be interpreted as Expectation-Maximization (EM) procedures, which enables Bayesian inference via Gibbs sampling. This construction yields tractable complete-data likelihoods and convergence guarantees for the associated Markov Chain Monte Carlo methods. From a broader evaluation perspective, Bayesian smoothing reduces sensitivity to sparse or imbalanced comparisons and provides uncertainty estimates for the inferred item skills.

Annotator-quality models address a complementary limitation: the assumption that every observed comparison is equally reliable. In crowdsourced or otherwise heterogeneous evaluations, participant quality can vary widely, motivating methods that explicitly account for annotator reliability (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates item quality and rater reliability through annotator-specific parameters. Their approach places Gaussian priors on item and rater parameters and uses likelihood scaling factors to modulate each annotator’s influence. Posterior inference is performed with gradient-based optimization, which can be difficult in high-dimensional settings because of local optima and because the updates do not provide convergence guarantees. Our EM-based method targets the same broad problem of heterogeneous rater reliability while providing guaranteed monotonic likelihood improvement. The Elo-based rater model developed by Google Research implements a simplified version of this approach. It is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and related research (Mentzer et al., 2020; Ballé et al., 2025).

To the best of our knowledge, no prior work combines the Bradley–Terry model with a Bayesian formulation that explicitly models rater quality while also providing closed-form EM updates. Relative to Chen et al. (2013), who use Gaussian priors and gradient-based optimization without convergence guarantees, our approach uses conjugate priors and an EM algorithm that ensures monotonic likelihood improvement. Relative to the Elo-based rater model widely used in CLIC (CLIC, 2025), which is a simplified heuristic relying on Elo-style updates, our method provides a principled Bayesian treatment with uncertainty estimates and interpretable rater-quality parameters that can be applied to pairwise-comparison benchmarks beyond a single domain.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

We study the general problem of converting noisy pairwise comparisons into a reliable ranking of items, a task that arises throughout machine learning, statistics, and human preference evaluation. Consider a set of K items that are repeatedly compared in pairs by a set of R raters. The observed dataset, denoted D , consists of the outcomes of these comparisons. For two items i and j , Bradley

& Terry (1952) proposed the following model:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is a parameter associated with item $k \in \{1, 2, \dots, K\}$ that represents its skill rating. This model gives an interpretable mechanism for inferring item scores from observed wins and losses across any collection of pairwise comparisons. Its basic form, however, assumes that all comparisons are equally reliable. Human evaluation datasets often violate this assumption because participants can differ in expertise, attention, diligence, hardware, or understanding of the task. Such differences create heterogeneous comparison noise that should be modeled rather than absorbed into item scores.

To represent varying rater reliability, we introduce a rater-specific quality parameter $q_r \in [0, 1]$. The parameter has a direct mixture interpretation: with probability q_r , rater r provides an informed judgment that follows the Bradley-Terry model. With probability $1 - q_r$, the same rater behaves as an uninformative chooser and guesses uniformly between the two items. This assumption yields the following mixture probability that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The log-likelihood of the data is given by:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ denotes the total number of comparisons between items i and j by rater r , and $w_{r,ij}$ is the number of times rater r ranks item i above item j . This likelihood does not admit a closed-form maximum likelihood estimator, so the optimal parameters λ and q cannot be written analytically. Generic iterative methods such as gradient descent could optimize the objective, but they introduce learning-rate choices and do not guarantee convergence. The Expectation–Maximization (EM) algorithm instead avoids learning-rate tuning while guaranteeing monotonic improvement of the observed-data likelihood and convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we use a Thurstonian interpretation of the Bradley-Terry model (Diaconis, 1988). Under this view, each comparison between items i and j is represented as a race whose random arrival times are Y_i and Y_j . The arrival times follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time wins the comparison. This construction recovers the standard Bradley-Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

To obtain tractable EM updates, we introduce latent variables that make the complete-data likelihood separable. First, define an indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which indicates whether the c -th comparison of items i and j by rater r follows the Bradley-Terry component of the mixture. Using these indicators, we define $Z_{r,ij}$ as the sum of the minimal arrival times over the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma latent-variable representation preserves the standard BT interpretation while making rater-specific reliability compatible with closed-form EM updates.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm computes maximum a posteriori (MAP) estimates for the item skills λ and rater qualities q by treating mixture assignments and Thurstonian arrival times as latent variables. Under the standard EM construction, the procedure converges to a stationary point of the posterior distribution.

We place priors on both parameter families. The item skills λ_k receive a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, conjugate to the exponential arrival-time model. Each rater quality parameter q_r receives a Beta prior, $q_r \sim B(\alpha, \beta)$, matching its interpretation as a probability.

The algorithm repeatedly maximizes the expected complete-data log-posterior conditioned on the current parameter estimates (λ^*, q^*) . It alternates between the following two steps until convergence:

E-step: Expectation In the E-step, we compute the expected complete-data log-posterior, denoted Q . This quantity is the expected log-posterior of the observed and latent variables under the current parameter estimates and the observed data. It is defined as:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A,Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood, ℓ_c , decomposes into three components.

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization The M-step updates the parameters by maximizing the Q function. Using the expectations from the E-step, this maximization reduces the original mixture objective to closed-form updates. The central quantity in both updates is the posterior probability that a particular comparison from rater r belongs to the Bradley-Terry component rather than the random-choice component. This probability, denoted γ , acts as a data-dependent weight for the comparison under the current model predictions. It is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j . Thus, γ quantifies how strongly the current model attributes a comparison to informed preference behavior rather than random choice. Higher γ values assign greater influence to the corresponding comparison.

The update for a rater's quality, q_r , has the form of a regularized weighted average. The numerator aggregates the rater's effective informed judgments, with each contribution weighted by the posterior probability (γ) that it came from the Bradley-Terry component, and then adds the Beta-prior contribution. The denominator normalizes by the rater's total number of comparisons and the prior mass. As a result, raters whose comparisons consistently align with the inferred preference structure receive higher quality estimates. The update is:

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The update for an item skill, λ_i , balances effective positive evidence against the amount of comparison exposure. The numerator sums the effective wins for item i across raters, again weighted

by γ , and includes the Gamma-prior contribution. The denominator accounts for the comparisons involving item i through the current skill estimates of its opponents, with the Gamma rate acting as regularization. The update is:

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a-1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R \left[w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)} \right]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

A full derivation of the Expectation-Maximization algorithm is provided in Appendix B.

4 EXPERIMENTAL RESULTS

We evaluate our **Bayesian Bradley-Terry** model with rater **Quality (BBQ)** against two BT-based baselines that represent common choices for pairwise-comparison aggregation: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012) and a gradient-descent BT model with rater quality (Crowd-BT) (Chen et al., 2013). For Crowd-BT, we use the Google Research implementation (Google, 2025), which was also used by CLIC (2025) and Ballé et al. (2025). The hyperparameters used in all experiments are reported in Section C.

The evaluation spans preference datasets from both language and image domains: large language model benchmarks (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)), image-compression studies (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), and a newly collected inhomogeneous rater quality (IHQ) dataset derived from the CLIC2024 image test set (CLIC, 2025). To isolate the role of rater quality within the same task and item set, we partition IHQ into screened and unscreened subsets. This design lets us compare aggregation behavior under high-quality, low-quality, and mixed rater populations without changing the underlying evaluation task. Details of the IHQ collection are given in Section E, and dataset statistics for comparisons, raters, and items are summarized in Section D and Table 2.

Real preference datasets do not provide external ground-truth rankings. We therefore first validate the models on simulated datasets, where all methods recover the same ordering as the number of samples increases. For the empirical datasets, we use the ranking obtained from the full dataset as a practical reference ordering. As a sanity check for this approximation, we examine whether the methods identify the same top-1 item on the full data. For every dataset except IHQ-unscreened, all three models select the same top-ranked item. On IHQ-unscreened, Crowd-BT and Bayes-BT do not identify the reference image as the highest-quality item, whereas BBQ does.

A useful aggregation method should produce similar rankings when a study is resampled or repeated under comparable conditions. We approximate this reproducibility with bootstrapping, which estimates ranking variability from the available rater data. The bootstrapping procedure is described in Section F. The WD and HiFiC datasets used active comparison-pair selection rather than purely random sampling. HiFiC used a binary-search strategy, while WD selected pairs by maximum information gain. Consequently, bootstrap estimates for these two datasets should be interpreted as stability diagnostics rather than fully independent-study replicates.

We report two complementary stability metrics. Top-1 agreement is the fraction of bootstrap samples whose best item matches the best item from the full dataset. Kendall’s Tau (τ) measures ordinal correlation between complete rankings (Kendall, 1938), with larger values indicating greater rank stability. Top-1 agreement emphasizes winner selection, whereas Kendall’s Tau evaluates the broader ordering across items.

4.1 PERFORMANCE ACROSS DATASETS

We compare the BT variants across datasets to assess ranking stability under different domains, dataset sizes, and rater populations. Table 1 summarizes Top-1 agreement and Kendall’s Tau. Datasets with more comparisons per model, such as MT-Bench, WD, and HiFiC, are generally more stable. HUMAINE deviates from this simple trend, indicating that ranking difficulty depends not only on the number of comparisons but also on factors such as rater consistency, item diversity, and comparison structure.

Table 1: Performance of BT-based aggregation methods across language and image preference datasets. Top: Top-1 agreement [%], Bottom: Kendall’s Tau. Across these evaluation conditions, BBQ most frequently identifies the top-performing item, obtains the best overall ranking stability on more than half of the datasets, and ranks second on the remaining datasets.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	<u>0.9245</u>	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

Across the evaluated datasets, BBQ provides the most consistently stable aggregation behavior. It identifies the top-performing item most often and is tied for best Top-1 agreement on three datasets. For complete rankings, it achieves the highest Kendall’s Tau on five of eight datasets and ranks second on the remaining three.

The three datasets where BBQ ranks second in Kendall’s Tau are settings with relatively high-quality and homogeneous raters. On MT-Bench, all models perfectly recover the top item. The WD dataset collected by Ballé et al. (2025) includes only five raters, likely the paper’s authors, suggesting a carefully controlled evaluation. In IHQ-screened, raters were explicitly filtered for quality. These cases are consistent with the model’s intended scope: when rater reliability is already uniform and high, explicitly modeling rater quality has less room to improve over simpler BT variants.

The IHQ subsets make the effect of rater heterogeneity especially clear. BBQ substantially outperforms Crowd-BT and Bayes-BT when low-quality raters are included, as in IHQ-all and IHQ-unscreened. All three models obtain their best Top-1 accuracy on the screened subset, reinforcing that rater selection remains valuable when it is feasible. Although Crowd-BT also models rater quality, its performance drops on the full dataset, likely because each crowdsourced rater contributes fewer than 40 comparisons, leaving limited evidence for reliable gradient-based rater-quality estimation. BBQ maintains strong performance without screening, with only a minor Top-1 decrease, indicating robustness in the heterogeneous-rater conditions targeted by the method.

Uniformly high rater quality is difficult to guarantee in many realistic evaluation pipelines. Large-scale crowdsourcing introduces variability, screening procedures add cost, and subjective criteria can affect even diligent annotators. In such settings, BBQ offers a principled way to retain information from partially reliable raters while reducing the influence of noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

As suggested by Table 1, datasets with more comparisons per model generally produce more stable rankings. Additional comparison evidence can come from two sources: adding raters or increasing the number of comparisons performed by each rater.

Figure 1 separates these two sources of evidence for the three BT variants. The left column varies the number of raters while fixing comparisons per rater at the maximum, and the right column varies comparisons per rater while fixing the number of raters at the maximum. Top-1 accuracy shows clear model differences, whereas Kendall’s τ is more similar across methods. Crowd-BT fails to converge with only one or two raters, underscoring the advantage of an EM procedure with convergence guarantees over gradient descent in sparse-rater regimes.

Crowd-BT and BBQ behave similarly when either the number of raters or the number of comparisons per rater is very small. This is expected because identifying rater quality requires repeated evidence across both raters and comparisons. As either source of evidence grows, BBQ increasingly

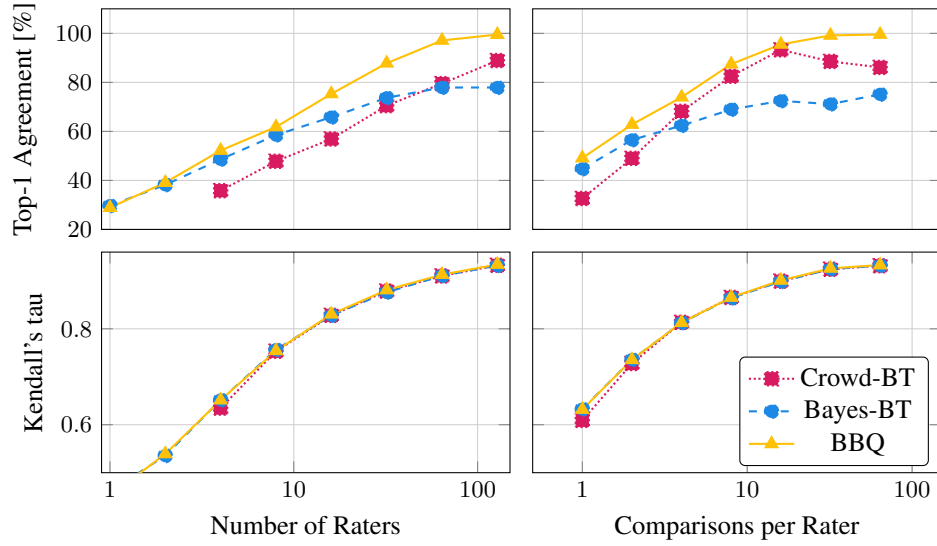


Figure 1: Scaling behavior of Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on the IHQ-all dataset. **Left:** Performance as the number of raters varies. **Right:** Performance as the number of comparisons per rater varies. Both Top-1 agreement and Kendall’s τ improve as more rater or comparison evidence is available. Top-1 agreement separates the methods more clearly than Kendall’s τ , which remains similar across models. Crowd-BT fails to converge with very few raters, illustrating the practical value of EM convergence guarantees. Crowd-BT and BBQ are similar in the sparsest regimes, while BBQ outperforms Bayes-BT as rater or comparison evidence increases.

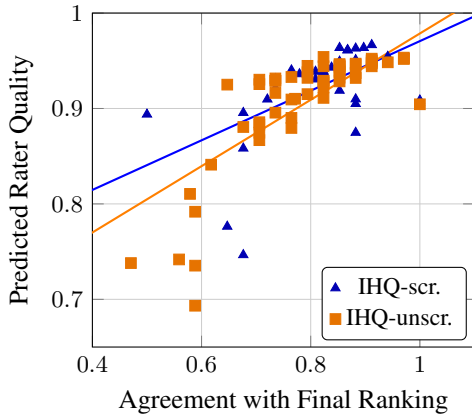


Figure 2: Relationship between rater agreement with the final ranking (x-axis) and predicted rater quality (y-axis) for the IHQ datasets. Each point represents one rater. Triangles denote the filtered dataset, and squares denote the unfiltered dataset.

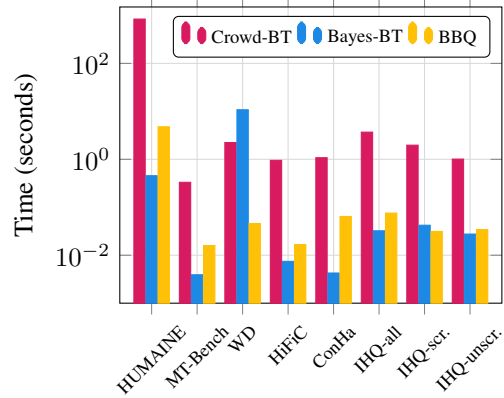


Figure 3: Average computation time in seconds (log-scale) for three models on one bootstrapped sample across eight datasets. Crowd-BT can be computationally expensive on some datasets, whereas BBQ remains fast across the evaluated conditions.

outperforms Bayes-BT. Bayes-BT generally underperforms overall, but it can surpass Crowd-BT when there are very few raters or limited per-rater data, because Crowd-BT cannot reliably estimate rater quality in such sparse settings.

4.3 RATER QUALITY

Figure 2 relates each rater’s agreement with the final ranking to the rater quality predicted by BBQ. The relationship is clearly positive: raters whose judgments align more closely with the consensus ranking receive higher estimated quality. For the filtered dataset, the Pearson correlation is $r = 0.551$ across 50 raters. For the unfiltered dataset, where rater reliability varies more widely, the correlation increases to $r = 0.724$ across 62 raters.

The unscreened subset contains several raters with both low agreement and low predicted quality, as expected for a more heterogeneous rater population. Some IHQ-unscreened raters fall below random-guessing agreement (50%), indicating systematic disagreement with the majority. This pattern illustrates why rater-quality modeling matters when pairwise comparisons are collected under less controlled conditions. BBQ identifies these raters and assigns them lower quality scores, reducing their influence on the final ranking while still using informative comparisons from the rest of the dataset.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average time required by each method to process one bootstrapped sample across datasets. These timings indicate whether the methods are practical for repeated resampling, large preference studies, and iterative evaluation workflows. BBQ converges within a few seconds on every evaluated dataset. Crowd-BT is slower across the board and is especially expensive on datasets with many comparisons, such as HUMAINE, where one convergence run takes around 15 minutes. Bayes-BT is slightly faster than BBQ on most datasets but converges slowly on WD. BBQ’s efficiency follows from the closed-form EM updates derived above, which avoid gradient tuning while remaining fast on the evaluated datasets.

This comparison is conservative with respect to implementation effort: Crowd-BT is highly optimized and implemented in C (Kernighan & Ritchie, 1988), whereas BBQ is implemented in plain NumPy without specialized optimization. A compiled or more aggressively vectorized implementation could therefore further reduce BBQ runtime. In the evaluated conditions, BBQ is practical not only for one-off aggregation but also for large preference studies and workflows that require repeated bootstrapping. The combination of stability, interpretability, and speed supports its use in realistic human-evaluation pipelines.

4.5 CONFIDENCE INTERVALS

We also test how Crowd-BT, Bayes-BT, and BBQ quantify uncertainty in item rankings. In a controlled simulation, raters compare equally skilled items under H_0 . Each model estimates skills and constructs uncertainty intervals used to test for significant differences, as detailed in Section G. The resulting Type I error rates are shown in Figure 6. At the 99% confidence level, the expected false-positive rate is 1%. Crowd-BT and BBQ are close to this target, with slightly lower rates, whereas Bayes-BT is more conservative at $\sim 0.1\%$. Thus, under this controlled equal-skill setting, Crowd-BT and BBQ provide well-calibrated uncertainty estimates, while Bayes-BT produces intervals that are conservative for significance testing.

5 LIMITATIONS

Although BBQ is designed for heterogeneous rater populations, its advantages naturally diminish when all raters are uniformly reliable. In such homogeneous datasets, explicit rater-quality modeling has little variability to exploit. Ensuring consistently high-quality raters, however, often requires substantial cost, screening, and control, which may not be feasible in large-scale studies. This tradeoff clarifies the method’s scope: BBQ is most useful when rater reliability is uncertain or mixed, and less beneficial when evaluations are already carefully curated.

A second limitation is the feedback format. BBQ currently applies to pairwise comparisons, while many evaluation protocols collect ratings, full or partial rankings, or multi-way choices. Extending the same rater-quality and Bayesian-aggregation ideas to these richer feedback types would broaden the range of applicable study designs.

Methodologically, our empirical conclusions are based on the available datasets and a finite number of bootstrap resamples. Bootstrapping provides a practical stability estimate, but larger and more varied studies would further test robustness across domains and sampling protocols. The model also assumes independence across comparisons and does not explicitly model context, presentation order, or fatigue effects, all of which can influence human judgments in practice.

Finally, although BBQ is fast in our experiments, extremely large item or rater sets may still require optimized, vectorized, or compiled implementations.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model for aggregating noisy pairwise comparisons while jointly estimating item quality and rater reliability. The core contribution is an Expectation-Maximization (EM) algorithm that estimates item skills and individual rater quality together, down-weighting unreliable comparisons without discarding the broader structure of the dataset. By explicitly modeling rater heterogeneity, the method produces more stable and accurate rankings in settings where rater reliability varies. The EM formulation provides rapid convergence and monotonic likelihood improvement, addressing an important limitation of gradient-based rater-quality approaches.

Across the evaluated language and image-compression preference datasets, BBQ achieves high Top-1 agreement and strong Kendall’s Tau, indicating robust ranking behavior across multiple domains and rater populations. The model is especially useful when raters are noisy, crowdsourced, or unevenly reliable, including cases where each rater contributes relatively few comparisons. These results support the broader principle that rater-quality modeling is most important when evaluation quality is heterogeneous or partially unreliable. The Bayesian framework also provides uncertainty estimates for item scores, enabling more interpretable comparisons across studies. Our controlled uncertainty experiment shows that BBQ’s error bars are well calibrated for testing whether equally skilled items are spuriously separated. Predicted rater quality aligns with agreement to final rankings, providing empirical evidence that the model identifies reliable evaluators in the IHQ setting.

Overall, BBQ advances human-evaluation methodology by offering a practical and interpretable approach for aggregating noisy pairwise comparisons under heterogeneous rater quality. Within the supported experimental conditions, it provides a path toward more cost-effective, interpretable, and reproducible studies for evaluating AI-generated content and other items ranked from human preferences.

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A USAGE OF LLMs

During the preparation of this paper, we used large language models (LLMs) as auxiliary writing and development tools. ChatGPT, Claude, Gemini, and Grammarly supported spellchecking, phrasing refinement, and text condensation for clarity and readability. ChatGPT, Claude, and Cursor also supported code analysis, completion, and visualization generation during the development workflow. All core research ideas, experimental design choices, and interpretations of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider the general setting of R raters comparing K items through pairwise judgments. Each rater r has quality q_r , meaning that with probability q_r they follow the Bradley–Terry model, and with probability $1 - q_r$ they choose randomly. The following EM (Dempster et al., 1977) derivation shows how these rater-specific reliabilities enter the Bayesian estimation framework.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley-Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley-Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley-Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley-Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

For the general pairwise-comparison setting with rater qualities, the complete-data objective is:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

The latent arrival-time contribution is:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

The rater-quality assignment contribution is:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

The observed comparison contribution is:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Combining these terms gives the complete-data log-likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We use conjugate priors so the same derivation applies across item sets and rater populations: a Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and a Beta distribution $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expected complete-data log-posterior under the current parameter values:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

The posterior probability that an annotation follows the Bradley-Terry component is:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

The expected sufficient statistics for the EM update are:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j,j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j,j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Substituting these expectations yields the expected complete-data log-posterior:

$$\begin{aligned}
\mathbb{E}[\ell_c] &= \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\
&+ \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&+ \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&+ \text{const.}
\end{aligned}$$

The priors add the following regularization terms:

$$\begin{aligned}
\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\
\mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].
\end{aligned}$$

Final Q -function:

$$\begin{aligned}
Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\
& + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
& \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
& + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
& + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] \\
& + \sum_{r=1}^R [(\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r)] \\
& + \text{const.}
\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function with respect to (λ, q) while holding the E-step expectations fixed, yielding updates that remain valid for any dataset represented by these sufficient statistics.

Update for q_r The update for each rater quality q_r follows by maximizing Q with respect to q_r , including the Beta prior that regularizes sparse per-rater evidence.

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
0 &\stackrel{!}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
\Rightarrow &\quad (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
\Rightarrow &\quad q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
\Rightarrow &\quad q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
\Rightarrow &\quad q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit rater-quality update at iteration t .

Update for λ_i The update for each item skill λ_i follows by maximizing Q with respect to λ_i , including the Gamma prior that regularizes item scores.

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^R \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^R \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^R \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit item-skill update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with its default hyperparameters.

For Bayes-BT and BBQ, the only hyperparameters are the prior parameters and stopping thresholds. For both models, we declare convergence when no ELO score changes by more than 1 between consecutive iterations. The ELO score is computed from the skill parameter λ as:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where we set ELO_SCALE_FACTOR to 400.

We use a Gamma prior with shape 5 and rate 0.1 for the skill parameters in both Bayes-BT and BBQ. For BBQ, the Beta prior on rater quality uses $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate the models on human preference datasets spanning language and image-compression domains. Table 2 summarizes the dataset scale, rater counts, item counts, and comparison density.

For natural language, we use HUMAINE (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which provides crowd-worker model comparison judgments on multi-turn dialogues.

For image compression, we consider WD (Ballé et al., 2025), a dense expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which focuses on conditional generative models. These datasets vary substantially in scale, rater count, comparison density, and rater expertise, providing complementary conditions for testing robustness.

We also introduce the inhomogeneous rater quality (**IHQ**) dataset, collected through a user study on CLIC2024 (CLIC, 2025) data via the *Mabyduck* platform. It contains two-alternative-forced-choice (2AFC) judgments across 28 generative image compression models. To study rater-quality effects while holding the task fixed, we provide two subsets: (i) **screened**, where raters passed pre-screening checks for attention and display quality, and (ii) **unscreened**, where all raters are included. The screened subset represents a higher-control evaluation condition, whereas the unscreened subset reflects noisier conditions common in large-scale human evaluations.

Together, these datasets let us examine aggregation behavior in large benchmarks, carefully controlled studies, and noisier settings where modeling rater reliability is especially relevant.

Table 2: Summary of datasets used in our experiments. The IHQ dataset was collected on the Mabyduck platform and includes both 2AFC and 3AFC settings. For IHQ, screened subsets include only raters who passed pre-screening tests for attention and display quality, whereas unscreened subsets include all raters and therefore represent a more heterogeneous rater population.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

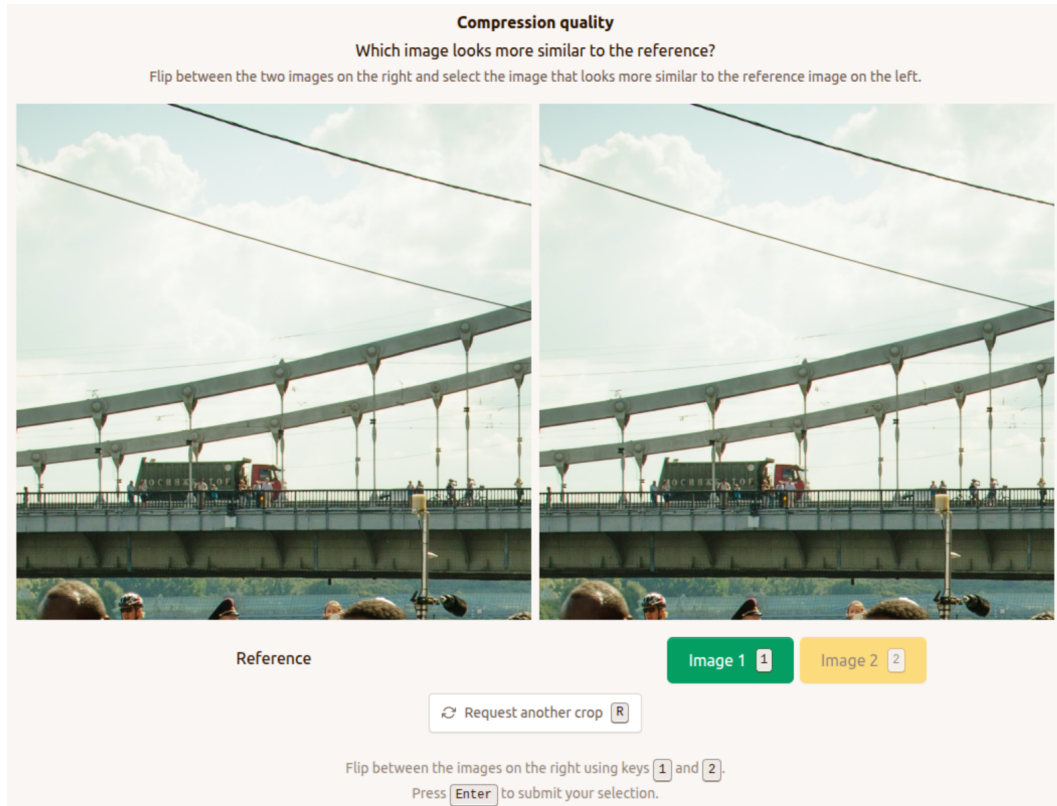


Figure 4: Screenshot of the Mabyduck user study platform used to collect pairwise image comparisons. The reference image appears on the left, and the rater selects between two compressed images on the right.

E USER STUDY PLATFORM

All pairwise comparisons for the CLIC2024 (CLIC, 2025) data were collected using the Mabyduck platform (Ltd., 2025). Figure 4 shows the interface used by raters. The task asks: “Which image looks more similar to the reference image?” Ties are not allowed, and all pairs were selected uniformly at random.

To study rater quality under matched task conditions, we split IHQ into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to sub-

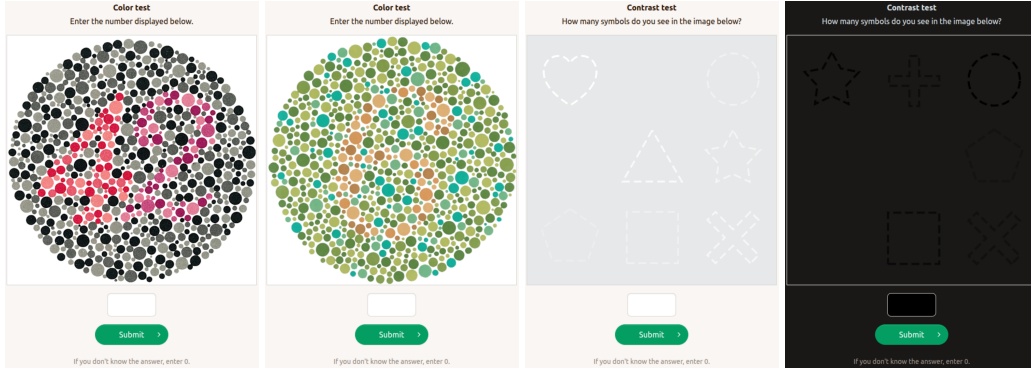


Figure 5: Four example images from the rater pre-screening process. The first two are color-blindness tests requiring raters to identify the number in each pattern. The last two are low-contrast shape-detection tests: one light gray shape on a white background and one dark gray shape on a black background.

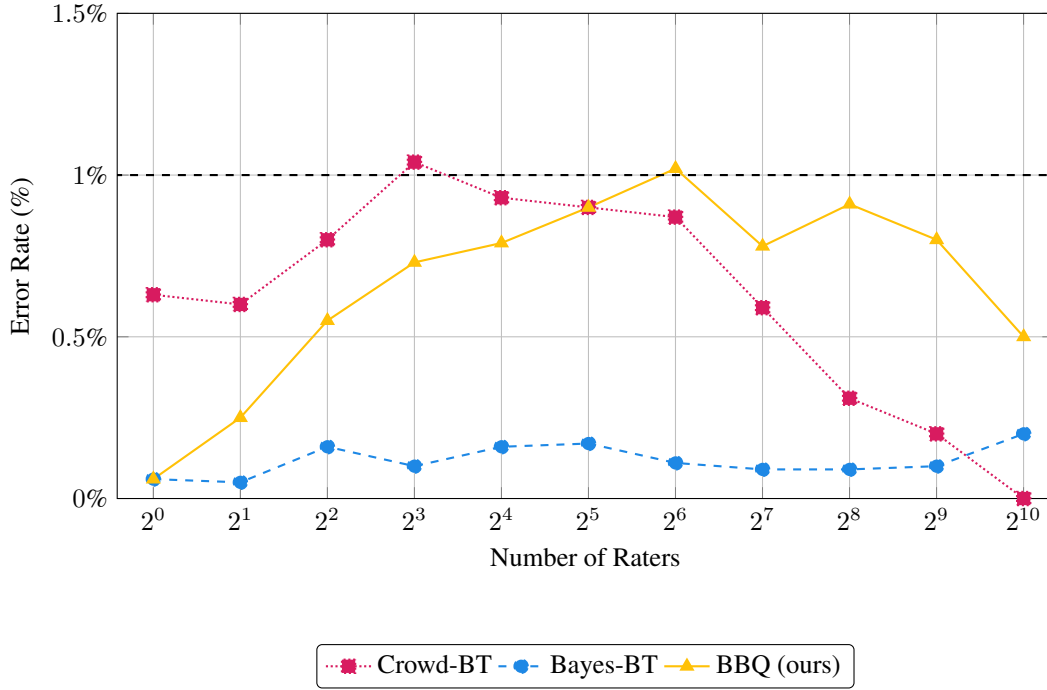


Figure 6: Type I error rate as a function of the number of raters in the equal-skill simulation. The dashed line indicates the expected 1% error rate at the 99% confidence level.

tle differences, and they completed training comparisons that reinforced the higher-quality-image selection criterion.

Figure 5 shows four example images from the pre-screening process. The first two are standard color-blindness tests requiring raters to identify the number in each pattern. The final two are shape-detection tests measuring sensitivity to low-contrast objects: one light gray shape on a white background and one dark gray shape on a black background. These checks make the screened subset a more controlled evaluation condition by limiting it to raters with adequate visual capabilities.

F BOOTSTRAPPING DETAILS

Because both BBQ and Crowd-BT explicitly model rater quality, we bootstrap over raters rather than over individual comparisons. This preserves each rater’s comparison distribution and provides a stability assessment aligned with the modeled data-generating units. For all datasets, we perform 10,000 bootstrap resamples of raters, except on HUMAINE, where one Crowd-BT bootstrap iteration takes approximately 15 minutes as discussed in Section 4.4. For HUMAINE, we therefore use 1,000 resamples to keep computation tractable.

G UNCERTAINTY ESTIMATION DETAILS

Beyond producing a point ranking, uncertainty estimation is needed to assess whether observed item differences are statistically meaningful.

Bayesian Bradley-Terry (Bayes-BT) and Bayesian Bradley-Terry with Quality (BBQ) quantify uncertainty through the posterior distribution over item skills. Each skill has a Gamma prior that is updated using the observed pairwise comparisons. Credible intervals from the posterior are converted to the Elo scale so item differences can be interpreted on the same scale as the reported rankings.

The non-Bayesian Crowd-BT model estimates uncertainty using a frequentist approximation. Specifically, it applies a second-order Taylor expansion around the maximum likelihood estimate. The Hessian of the log-likelihood is inverted to obtain a covariance matrix, whose diagonal entries correspond to item-level variances. These variances are converted to 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the constants scale the standard deviation to the 99% confidence level. Sharper likelihood peaks yield smaller intervals and higher certainty, while flatter peaks yield larger intervals and greater uncertainty.

To evaluate uncertainty calibration, we run a controlled simulation. Each trial contains two equally skilled items with a 50–50 win probability, and pairwise comparison outcomes are simulated with coin flips. For each trial, we simulate multiple raters and apply each model to estimate Elo scores and associated uncertainty.

Formally, the null hypothesis H_0 states that the two items are equally strong. For each trial, we compute the 99% confidence or credible interval for each item skill estimate. A model incorrectly rejects H_0 if the intervals do not overlap, which would indicate a statistically significant item difference when none exists. The primary metric is the frequency of these incorrect rejections, i.e., the observed Type I error rate.

We perform 10,000 trials with 50 comparisons per rater while varying the number of raters, allowing us to study how uncertainty estimates scale with data volume. This setup estimates each model’s empirical Type I error rate and compares it with the theoretical 1% expectation at the 99% confidence level. As shown in Figure 6, Crowd-BT and BBQ have Type I error rates close to 1%, indicating good calibration in this equal-skill simulation. Bayes-BT is more conservative, consistently producing lower error rates than expected, which suggests its credible intervals are less calibrated for this significance-testing use case.